

$^{32}\text{S}(\alpha, p\gamma)$ 1970Ho09, 1972Br33, 1973Al22

$J^\pi=0^+$ for ^{32}S ground state.

- 1970Ho09:** 10.975-, 11.960-, 12.120-MeV α beams were produced from the Chalk River MP tandem accelerator. Targets were 100- $\mu\text{g}/\text{cm}^2$ Sb_2S_3 on a 10- $\mu\text{g}/\text{cm}^2$ carbon backing. γ rays were detected using 15.2-cm-long by 12.7-cm-diam NaI(Tl) detectors. Protons were detected using an annular silicon surface detector close to 180° with FWHM=30 keV for protons and FWHM=50 keV for α . Measured E_p , E_γ , I_γ , $p\gamma(\theta)$. Deduced 25 level energies from E_p , J , branching ratios, multipole mixing ratios.
- 1972Br33:** a 10-MeV α beam was produced from the Chalk River MP tandem accelerator. Targets were 340- $\mu\text{g}/\text{cm}^2$ Sb_2S_3 on gold backings. γ rays were detected using a 44-cm³ Ge(Li) detector inside an anticoincidence split annular NaI(Tl) detector. Measured E_γ , I_γ . Deduced 11 level energies, 16 γ branching ratios, 11 lifetimes using the Doppler Shift Attenuation Method (DSAM). **1970In01** from the same experiment deduced the lifetime using DSAM for a level at 3942 keV and the mixing ratio for a 3942- $\rightarrow$ 2646 γ transition.
- 1973Al22:** a 14-MeV α beam was produced from the 80-cm cyclotron at cyclotron at the Research Institute for Physics in Stockholm and 12-16-MeV α beams were produced from the Uppsala tandem accelerator. Targets were prepared by depositing pulverized sulphur enriched to 99.9% in ^{32}S on lead backings. γ rays were detected using an Ortec 60-cm³ Ge(Li) detector with FWHM=2.2 keV at 1333 keV and an additional Canberra 60-cm³ Ge(Li) detector for $\gamma\gamma$ -coincidence measurements. Measured E_γ , I_γ , $\gamma\gamma$ -coin, 4 $\gamma(\theta)$. Deduced 12 level energies from E_γ , J for 4 levels, branching ratios, and mixing ratios.
- 1969In04:** 6-8-MeV α beams were produced from the Chalk River MP tandem accelerator. A thick target (270 $\mu\text{g}/\text{cm}^2$) of natural sulfur on a tantalum backing and a thin target (50 $\mu\text{g}/\text{cm}^2$) of Sb_2S_3 on a gold backing were used. γ rays were detected using two Ge(Li) detectors of 15 and 40 cm³ volume. Measured E_γ , I_γ . Deduced 6 level energies, branching ratios, 5 lifetimes using the Doppler Shift Attenuation Method (DSAM) and 1 lifetime for the 3163-keV level using the Recoil Distance Method (RDM), and transition strengths.
- 1974Lo17:** 11.0, 12.0 and 14.7-MeV α beams were produced from the Oliver Lodge Laboratory at the University of Liverpool. Targets were natural sulphur in CdS. Protons were detected using an annular ΔE -E telescope close to 180° with FWHM=80 keV. γ rays were detected using five NaI(Tl) and one Ge(Li) detectors. Measured E_γ , I_γ , $p\gamma$ -coin, $\gamma\gamma$ -coin, γ -ray linear polarization. Deduced J and π for 3 levels, 6 γ branching ratios, 5 mixing ratios, and lifetimes for a 5408-keV level using DSAM and for a 6088-keV level using RDM.
- 1973Br26:** a 11.2-MeV α beam was produced from the Legnaro (Padova) Van de Graaff generator. Targets were natural CdS. γ rays were detected using a Ge(Li) detector. Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.
- 1973An01:** 5.5- and 8.6-MeV α beams of 100-200 nA were produced from the Oxford Vertical Van de Graaff (Injector) or the Oxford E(n) Tandem. Targets were natural sulphur in AgS. γ rays were detected using a Ge(Li) detector at 0° . Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

 ^{35}Cl Levels

E(level) [†]	J^π	$T_{1/2}$	Comments
0	$3/2^+$		
1219.5 10	$1/2^+$	0.21 ps $+10-8$	$T_{1/2}$: from 1969In04 using DSAM.
1763.1 8	$5/2^+$	0.55 ps 15	J^π : spin=5/2 from $\gamma(\theta)$ in 1970Ho09 . Parity is from the Adopted Levels. $T_{1/2}$: from 1969In04 using DSAM.
2645.6 10	$7/2^+$	0.30 ps 9	J^π : spin=7/2 from $\gamma(\theta)$ in 1970Ho09 . Parity is from the Adopted Levels. $T_{1/2}$: from 1969In04 using DSAM.
2693.5 13	$3/2, 5/2$	70 fs 30	J^π : spin=3/2, 5/2 from $\gamma(\theta)$ in 1970Ho09 . $T_{1/2}$: from 1972Br33 using DSAM. Other: $\tau < 30$ fs (1969In04 , DSAM).
3002.1 15	$5/2$	48 fs 9	$T_{1/2}$: from 1972Br33 using DSAM. Other: $\tau < 50$ fs (1969In04 , DSAM). J^π : spin=3/2, 5/2 from $\gamma(\theta)$ in 1970Ho09 . 1970Ho09 states that 3/2 causes $M2 > 200$ W.u. or $E2 > 10$ W.u and is ruled out.
3163.0 11	$7/2$	60 ps 7	J^π : spin=7/2 from $\gamma(\theta)$ in 1970Ho09 . $T_{1/2}$: from 1969In04 .
3915 2		<22 fs	E(level): from 1972Br33 . $T_{1/2}$: from 1972Br33 using DSAM.
3943.2 13		330 fs 50	$T_{1/2}$: from 1972Br33 using DSAM.
3968 2		<54 fs	E(level): from 1972Br33 . $T_{1/2}$: from 1972Br33 using DSAM.
4058 3		31 fs 13	E(level): weighted average of 4045 10 (1970Ho09) and 4058 2 (1972Br33).

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2}	Comments
4108 2	3/2, 7/2	70 fs 16	T _{1/2} : from 1972Br33 using DSAM. E(level): weighted average of 4110 10 (1970Ho09) and 4108 2 (1972Br33). J ^π : from γ(θ) in 1970Ho09.
4171 2		68 fs 24	T _{1/2} : from 1972Br33 using DSAM. E(level): weighted average of 4175 10 (1970Ho09) and 4171 2 (1972Br33). T _{1/2} : from 1972Br33 using DSAM.
4347.7 13	9/2 ⁻	2.9 ps +14-7	J ^π : from γ(θ) and POL in 1974Lo17. Other: J=5/2, 9/2 from γ(θ) in 1970Ho09 and 1973Al22.
4618 2	3/2, 5/2	58 fs 24	T _{1/2} : from 1972Br33 using DSAM. E(level): from 1972Br33. J ^π : from γ(θ) in 1970Ho09.
4770 15			T _{1/2} : from 1972Br33 using DSAM. E(level): from 1970Ho09.
4880.7 22		280 fs 60	E(level): other: 4882 2 from 1972Br33, 4885 20 from 1970Ho09. T _{1/2} : from 1972Br33 using DSAM.
5015 20			E(level): from 1970Ho09.
5175 20			E(level): from 1970Ho09.
5230 20	1/2, 3/2, 5/2		E(level): from 1970Ho09.
5407.0 15	11/2	0.4 ps 1	J ^π : from γ(θ) in 1970Ho09. E(level): other: 5415 20 from 1970Ho09.
			J ^π : from γ(θ) and POL in 1974Lo17. Other: J=11/2 from γ(θ) and yield function in 1973Al22.
5535 20			T _{1/2} : from 1974Lo17 using DSAM.
5600 20			E(level): from 1970Ho09.
5650 20			E(level): from 1970Ho09.
5850 25	5/2, 9/2		E(level): from 1970Ho09.
5927.0 20	7/2, 11/2		J ^π : from γ(θ) in 1970Ho09. E(level): other: 5930 25 from 1970Ho09.
6086.9 18	13/2 ⁻	9.3 ps 8	J ^π : from γ(θ) in 1973Al22. J ^π : from γ(θ) and POL in 1974Lo17. Other: J=9/2, 13/2 from γ(θ) in 1973Al22.
6400 25			T _{1/2} : from 1974Lo17 using RDM. E(level): from 1970Ho09.

[†] From a least-squares fit to E_γ values with uncertainties, unless otherwise noted. 1969In04 and 1972Br33 reported E(level) with uncertainties from measured E_γ but did not report E_γ. 1970Ho09 reported E(level) with uncertainties from measured E_p. 1973Al22 is the only literature in this dataset that reported measured E_γ with uncertainties.

 $\gamma(^{35}\text{Cl})$

E _i (level)	J _i ^π	E _γ [†]	I _γ	E _f	J _f ^π	δ	Comments
1219.5	1/2 ⁺	1219.5 10	100	0	3/2 ⁺		E _γ : from 1973Al22.
1763.1	5/2 ⁺	543.6	<0.5	1219.5	1/2 ⁺		I _γ : from 1969In04.
		1763.1 10	100	0	3/2 ⁺	+3.0 1	E _γ : from 1973Al22.
							I _γ : from 1969In04, 1970Ho09, 1973Al22.
							δ: from 1970Ho09.
2645.6	7/2 ⁺	882.4 10	11.1 22	1763.1	5/2 ⁺	+0.25 5	E _γ : from 1973Al22.
							I _γ : from 1970Ho09. Other: 18 (1969In04), 10 (1973Al22).
							δ: from 1970Ho09.
		1426.1	<1.2	1219.5	1/2 ⁺		I _γ : from 1969In04.
		2646.0 20	100.0 22	0	3/2 ⁺		E _γ : from 1973Al22.
							I _γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	δ	Comments
2693.5	3/2, 5/2	930.5 15	18.2 26	1763.1	5/2 ⁺		E_γ : from 1973Al22. I_γ : from 1970Ho09. I_γ : from 1970Ho09. Other: 10 (1969In04). E_γ : from 1973Al22. I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). δ : +0.17 8 for J=3/2; -0.26 5 for J=5/2 (1970Ho09).
		1474.0	11.7 26	1219.5	1/2 ⁺		
		2693.2 20	100 4	0	3/2 ⁺		
3002.1	5/2	308.6	<5	2693.5	3/2, 5/2		I_γ : from 1969In04.
		356.5	<1	2645.6	7/2 ⁺		I_γ : from 1969In04.
		1239.0	<2	1763.1	5/2 ⁺		I_γ : from 1969In04. Other: <4 (1972Br33).
		1782.6	<2	1219.5	1/2 ⁺		I_γ : from 1969In04, 1972Br33.
		3002.0 15	100	0	3/2 ⁺		E_γ : from 1973Al22. I_γ : from 1969In04, 1970Ho09, 1972Br33, 1973Al22.
3163.0	7/2	160.9	<0.54	3002.1	5/2		I_γ : from 1969In04.
		469.5	<0.54	2693.5	3/2, 5/2		I_γ : from 1969In04.
		517.4 15	11.1 33	2645.6	7/2 ⁺		E_γ : from 1973Al22. I_γ : from 1970Ho09. Other: 8.7 (1969In04), 11 (1973Al22).
		1399.9	<1.1	1763.1	5/2 ⁺		I_γ : from 1969In04.
		3162.6 15	100.0 33	0	3/2 ⁺	-0.14 2	E_γ : from 1973Al22. I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).
3915		2695	19 5	1219.5	1/2 ⁺		δ : from 1970Ho09. I_γ : from 1972Br33.
		3915	100 5	0	3/2 ⁺		I_γ : from 1972Br33.
3943.2		1297.4 15	8.1 21	2645.6	7/2 ⁺		E_γ : from 1973Al22. I_γ : weighted average of 9.9 22 (1970Ho09) and 6.4 21 (1972Br33). Other: 11 (1973Al22).
		2180.2 15	100.0 21	1763.1	5/2 ⁺		E_γ : from 1973Al22. I_γ : from 1972Br33. Other: 100.0 22 (1970Ho09), 100 (1973Al22).
3968		2748	100	1219.5	1/2 ⁺		I_γ : from 1972Br33.
		3968	<2	0	3/2 ⁺		I_γ : from 1972Br33.
4058		2295	8 4	1763.1	5/2 ⁺		I_γ : from 1972Br33.
		2838	100 4	1219.5	1/2 ⁺		I_γ : from 1972Br33.
		4058	<2.2	0	3/2 ⁺		I_γ : from 1972Br33.
4108	3/2, 7/2	2345	16 5	1763.1	5/2 ⁺		
		4108	100 5	0	3/2 ⁺		
4347.7	9/2 ⁻	1184.6 10	100.0 31	3163.0	7/2	-0.36 3	E_γ : from 1973Al22. I_γ : from 1974Lo17. Others: 100 7 (1970Ho09) and 100 4 (1972Br33). δ : from 1974Lo17. Other: -0.40 +8-9 for J=9/2 (1970Ho09). A_2 =-0.95 3, A_4 =0.08 3 (1974Lo17). POL =-0.005 40 (1974Lo17).
		1702.1 15	52 7	2645.6	7/2 ⁺	-0.00 3	E_γ : from 1973Al22. I_γ : unweighted average of 61 7 (1970Ho09), 39 4 (1972Br33), and 56.3 31 (1974Lo17). δ : from 1974Lo17. A_2 =-0.34 5, A_4 =0.04 5 (1974Lo17). POL =0.36 7 (1974Lo17).
4618	3/2, 5/2	4618	100	0	3/2 ⁺		I_γ : from 1972Br33.
4880.7		2235.0	52 9	2645.6	7/2 ⁺		δ : +0.08 14 for J=3/2; -0.36 15 for J=5/2 (1970Ho09). I_γ : from 1970Ho09.
		3117.4 20	100 9	1763.1	5/2 ⁺		E_γ : from 1973Al22. I_γ : from 1970Ho09. Other: 100 (1973Al22).

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$^{32}\text{S}(\alpha, p\gamma)$ [1970Ho09](#), [1972Br33](#), [1973Al22](#) (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	δ	Comments
5230	1/2, 3/2, 5/2	5230	100	0	3/2 ⁺		I_γ : from 1970Ho09 .
5407.0	11/2	1059.3 10	9.9 22	4347.7	9/2 ⁻		E_γ : from 1973Al22 .
							I_γ : from 1974Lo17 .
							$A_2=0.19$ 11, $A_4=-0.01$ 13 (1974Lo17).
							$\text{POL}=-0.40$ 14 (1974Lo17).
		2244 3	100.0 22	3163.0	7/2		E_γ : from 1973Al22 .
							I_γ : from 1974Lo17 .
							$A_2=0.44$ 3, $A_4=-0.19$ 3 (1974Lo17).
5850	5/2, 9/2	2687	100	3163.0	7/2		δ : -2.2 5 or -0.6 1 for J=5/2; +0.28 4 for J=9/2 (1970Ho09).
5927.0	7/2, 11/2	1579.3 15	100	4347.7	9/2 ⁻		E_γ : from 1973Al22 .
							δ : +1.1 5 for J=7/2; -0.8 4 for J=11/2 (1973Al22).
6086.9	13/2 ⁻	679.9 10	100 6	5407.0	11/2	+0.020 15	E_γ : from 1973Al22 .
							I_γ : from 1974Lo17 .
							δ : from 1974Lo17 . Other: -0.00 7 for J=13/2 from 1973Al22 .
							$A_2=-0.22$ 1, $A_4=0.005$ 15 (1974Lo17).
							$\text{POL}=-0.45$ 5 (1974Lo17).
		1739.2	18 6	4347.7	9/2 ⁻		I_γ : from 1974Lo17 .

[†] Values with uncertainties are from [1973Al22](#) and E_γ values without uncertainties are from level-energy difference.

$^{32}\text{S}(\alpha, p\gamma)$ 1970Ho09, 1972Br33, 1973Al22

Level Scheme

Intensities: Relative photon branching from each level

